

Question	Marking guidance	Mark	Comments
02.1	Use of $\theta = \frac{\text{arc length}}{r}$ eg $\frac{12}{17} = 0.71$ ✓	1	
02.2	Use of $\omega = \frac{\theta}{t}$ (with any time from graph) OR use of $t = 0.12$ ✓ 5.8 or 5.9 (rad s ⁻¹) ✓	2	
02.3	Idea that sheet doesn't clear light gate (after 18 swings) ✓	1	
02.4	Max 3 from: ✓✓✓ Change in flux/flux linkage (in sheet) induces an emf (Which) induces a current Description of Lenz's law or that a force opposes the motion Force acting on a current-carrying conductor OR Reference to I^2R losses Energy losses come from E_k of sheet	3	Allow "cuts magnetic field lines" or change in magnetic flux density. Condone minor discrepancies in description of Lenz's law

Question	Marking guidance	Mark	Comments
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10	D
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11	D
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12	C
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Question	Marking guidance	Additional comments/guidelines	Mark
01.1	Two from: ✓ ✓ measure from equilibrium position time for ≥ 10 s time for ≥ 10 oscillations repeat timings and find mean		2
01.2	(resultant) force acts in opposite direction to displacement/x ✓	Allow 'to the left'	1
01.3	$ma = (-)kx$ ✓ Use of $a = (-)\omega^2 x$ ✓ Use of $\omega = 2\pi f$ and $T = \frac{1}{f}$ ✓ Correct handling of negative signs and power to give $T = 2\pi\sqrt{\frac{m}{k}}$ ✓	Don't allow $a = \omega^2 r$	4
Total			7

Question	Marking guidance	Additional comments/guidelines	Mark
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8	A
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9	D
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10	C
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Question	Answers	Additional comments/Guidelines	Mark	AO
02.1	Use of $F = kx$ and $F = mg$ seen or implied to give 521 (N m ⁻¹) ✓	note: full credit for 2.1 and 2.2 can be given if candidates attempt them in reverse order 519 (Nm ⁻¹) if a rounded mg used at least 3sf	1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
02.2	Use of $T = 2\pi\sqrt{\frac{m}{k}}$ to give 0.80 (s) ✓	alternative $T = 2\pi\sqrt{\frac{x}{g}}$ but not pendulum equation condone 1sf	1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
02.3	Use of $\omega = \frac{2\pi}{T}$ ($= 7.85 \text{ rad s}^{-1}$) ✓ Use of $E_{\text{oscillator}} = \frac{1}{2}m\omega^2 A^2$ ✓ 0.42 (J) ✓ Alternative: use of $\frac{1}{2}kx_2^2 - \frac{1}{2}kx_1^2$ to calculate change in EPE: $\frac{1}{2}k(0.20^2 - 0.16^2)$ ($= 3.75 \text{ J}$) ✓ calculates change in GPE correctly $= -3.34 \text{ (J)}$ ✓ 0.41 (J) ✓	full marks can be awarded for: idea that energy of oscillator = max potential energy ✓ $E_{P \text{ max}} = \frac{1}{2}kA^2$ ✓ 0.42 (J) ✓ accept 0.41 J or 0.42 J for full credit	3	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
02.4	Suitable axes scales and labels ✓ Sinusoidal shape (judged by eye) that is never below the x-axis ✓ Time period (of their graph) is 0.4 s ✓ First peak kinetic energy matches candidate's 02.3 answer (eg 0.42 J) ✓	suitable scale means intervals annotated and going in 2s, 4s, etc. sinusoidal shape includes that the steepest part passes through the correct point, and peaks and troughs are curved ignore amplitudes decreasing allow any starting value of KE	4	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
02.5	Any two: idea that force decreases (with subsequent cycles) ✓ idea that force varies within each cycle ✓ because damping force decreases with decreasing speed (or reverse argument)✓		2	AO3
Total			11	

Question	Answers	Additional comments/Guidelines	Mark	AO
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10	B
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21	D
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Question	Answers	Additional comments/Guidelines	Mark	AO
01.1	Evidence of use of $\omega = \frac{2\pi}{T}$ ✓ (to determine ω) Maximum kinetic energy read from graph ✓ Evidence of use of $E_k = \frac{1}{2} m v^2$ And $v_{\max} = \omega A$ ✓ To give $A = 25 \text{ mm}$ ✓	Expect to see $\frac{2\pi}{0.68}$ or 9.2 rad s^{-1} Expect an answer that rounds to $9.4 (\times 10^{-4} \text{ J})$ For MP3 the substitution or correct evaluation must be seen Expect to see $0.23 (\text{m s}^{-1})$ Give full credit for combined equations	4	1 × AO1 3 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
01.2	<u>Horizontal</u> line between $\pm \frac{A}{2}$ ✓ at 0.235 mJ ✓	If MP1 is not given, MP2 can be given if the maximum of the line is at 0.235 mJ Allow answer greater than 0.22 and less than or equal to 0.24	2	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
01.3	Use of $mg = ke$ to determine k, or seen in formula for T AND Use of $T = 2\pi\sqrt{\frac{m}{k}}$ ✓ to give $T = 0.41 \text{ s}$ ✓	Expect to see $k = 8.2 \text{ (N m}^{-1}\text{)}$ Accept use of $T = 2\pi\sqrt{\frac{e}{g}}$ provided e is explicitly the extension At least 2 sf	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
01.4	Use of $\omega = \frac{2\pi}{T}$ or seen in formula for v ✓ Use of $v = \pm \omega \sqrt{A^2 - x^2}$ With $A = 5.0 \times 10^{-2} \text{ (m)}$ or $x = 4.2 \times 10^{-2} \text{ (m)}$ ✓ To give $v = 0.41 \text{ (m s}^{-1}\text{)}$ ✓ OR Alternative calculation using energy: calculation of EPE at bottom ($1.58 \times 10^{-2} \text{ J}$) ✓ calculation of GPE at top ($3.16 \times 10^{-2} \text{ J}$) ✓ calculation of v from difference (KE) ✓	Expect to see $\omega = 15.3 \text{ (rad s}^{-1}\text{)}$ Or 15.7 if 0.4 used for T If $x = 0$ seen, award max 2 Allow POT error in substitution Accept 0.42(6) from rounding Give full credit for use of $x = A\cos(\omega t)$ (MP2) And $v = A\omega\cos(\omega t)$ and answer (MP3)	3	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
01.5	(After extension becomes zero) force is constant, so acceleration is constant (so no longer SHM) ✓ acceleration is no longer proportional to (-) displacement ✓	Alternatives for MP1 - Force (on the load) is equal to the weight - The acceleration (of the load) is g Treat suggestion that acceleration no longer opposite to displacement as neutral	2	AO4

Question	Answers	Additional comments/Guidelines	Mark	AO
01.6	Period of X is greater than period of Y and mass is the same ✓ discussion involving $T = 2\pi\sqrt{\frac{m}{k}}$ So stiffness of Y must be greater. ✓ OR Calculation of one stiffness. ✓ Calculation of other stiffness and correct comparison. ✓	Answer suggesting X has greater stiffness gets zero marks. Relationship between T, m, k may be expressed qualitatively For X stiffness = $3.0 \text{ (Nm}^{-1}\text{)}$ For Y stiffness = $8.2 \text{ (Nm}^{-1}\text{)}$	2	AO2
Total			15	

Question	Answers	Additional comments/Guidelines	Mark	AO
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17	B	At time $\frac{T}{2}$ the potential energy E_p of the particle is a minimum.	AO1
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09	D	$4k$ $16E$
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10	c	$g \sin \theta$
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Question	Answers	Additional comments/Guidelines	Mark	AO
02.1	Graph is straight line AND line goes through origin acceleration = constant $\times$ displacement OR acceleration and displacement are directly proportional ✓ Graph has negative gradient idea that the constant of proportionality is negative OR acceleration and displacement are in opposite directions ✓	Condone missing 'directly'	2	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
02.2	Value of acceleration and displacement read from graph ✓ Use of $a = -\omega^2 x$ and $\omega = 2\pi f$ OR $a = -(2\pi f)^2 x$ ✓ To give $f = 660$ (Hz) ✓	Accept any point on line or gradient calculation Eg displacement = 1×10^{-3} (m) and acceleration = -17.2×10^3 (m s ⁻²) Do not condone PoT error in MP1 Expect to see $4.2 (\times 10^3)$ for ω Accept answers that round to 660 or 670	3	1 × AO1 1 × AO2 1 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
02.3	Use of $v_{\max} = A\omega$ OR $v_{\max} = 2\pi f A$ with $A = 1$ mm OR their ω OR their f or 700 (Hz) ✓ to give $v = 4.2$ m s ⁻¹ ✓	Condone PoT error in MP1 Look for $\omega = 4.2 \times 10^3$ rad s ⁻¹ Use of show that value of 700 Hz gives 4.4	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
02.4	P at (0,0) ✓	Answer must be marked on graph	1	AO3
Total			8	

Question	Answers	Additional comments/Guidelines	Mark	AO
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8	C	1.09 m	AO1
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9	B	<i>E</i> <i>E</i>	AO1
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21	B	0.6 m	AO2
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22	A	$\left(\frac{1}{f\Delta L}\right)^2$	AO2
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Question	Answers	Additional comments/Guidelines	Mark	AO
02.1	Idea that when y is a maximum all of the energy is (gravitational) potential energy ✓ Idea max value of y is constant (so same total energy) ✓		2	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
02.2	Reads off maximum y and relates it to $GPE = mgy$ $_1$ ✓ Relates max GPE to max KE $_2$ ✓ 1.08 to 1.1 (m s^{-1}) $_3$ ✓	0.061 m. Accept 0.060 to 0.062 . Reject 0.06 seen in working. For MP1 and MP2, allow alternative use of amplitude and shm e.g.: Reads off max y and relates it to max angle = $\cos^{-1}((1.8 - y) \div 1.8)$ (= 15.0°) $_1$ ✓ Finds amplitude (0.47 m) and ω and uses $v = \omega A$ $_2$ ✓ If not other mark awarded, allow 1 MAX for use of $v = \omega A$ with a calculated ω and $A = 0.061$	3	AO3 $\times$ 1 AO2 $\times$ 2

Question	Answers	Additional comments/Guidelines	Mark	AO
02.3	Determines period of shorter swing using Figure 5 ✓ Use of <i>their</i> $T = 2\pi \sqrt{\frac{l}{g}}$ ✓ $l = 0.72$ to 0.56 OR idea that the length of shorter pendulum = $1.8 - h$ ✓ 1.08 to 1.24 m ✓	Accept 1.5s to 1.7s Use of means by correct substitution or rearrangement.	4	AO3 × 2 AO2 × 2
Total			9	

Question	Answers	Additional comments/Guidelines	Mark	AO
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08	A	maximum	zero	AO1

09	C	$2\pi\sqrt{\frac{M\Delta l}{(M+m)g}}$	AO2
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21	A	E 0	AO1
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Question	Answers	Additional comments/Guidelines	Mark	AO
01.1	Use of trig approach ✓ 4.9(4) (m) ✓	eg $2 \times 67.7 \times \sin(2.09)$ Condone $67.7 \times \sin(4.18)$ alternative converting to radians ✓ use of $d \approx r\theta$ to give answer ✓ accept 4.93 (m)	2	AO2

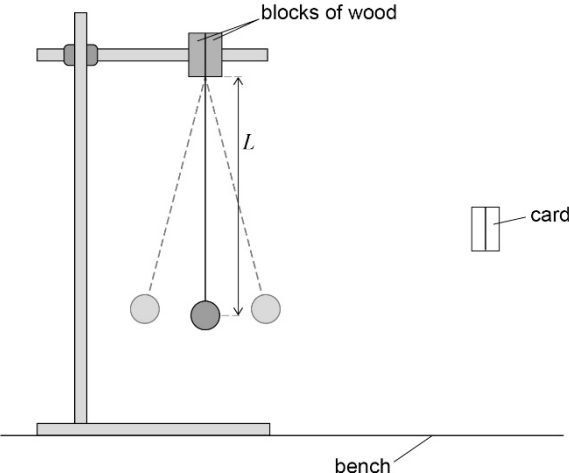
Question	Answers	Additional comments/Guidelines	Mark	AO
01.2	Use of $T = 2\pi\sqrt{\frac{l}{g}}$ and $T = \frac{2\pi}{\omega}$ ✓ Use of $\frac{1}{2}m\omega^2 A^2 = \frac{1}{2}mv^2$ OR $v = \text{their } A \times \text{their } \omega$ ✓ 0.94 (m s⁻¹) ✓	“Use of” means by substitution or rearrangement Look for $T = 16.5$ s For mp1 look for $\omega = \sqrt{\frac{g}{l}}$ Allow ecf for their 1.1 . Do not accept 67.7 m for their A. Note $A = 4.9$ gives 0.93 ms⁻¹ Alternative MP1 and MP2: Any two from: ✓✓ • $67.7 \times \cos(2.09)$ seen • height above ground = 67.7 (m) – calculated length seen • attempt to use $v = \sqrt{2gh}$ OR $mgh = \frac{1}{2}mv^2$	3	AO2

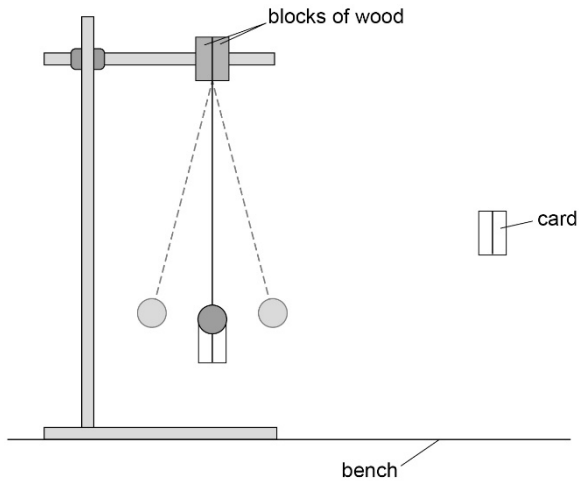
Question	Answers	Additional comments/Guidelines	Mark	AO
01.3	evidence of determination of circumference of path OR use of their path length $\div$ number of seconds in one day ✓ 304 (m s ⁻¹) ✓	Expect to see 2.63×10^7 (m) For MP1 accept alternative using $\omega = \frac{2\pi}{T}$ OR $v = r\omega$ with their ω Accept 305 or 300 (m s ⁻¹)	2	1 × AO1 1 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
01.4	Two from: ✓✓ • calculation of period (16.5 (s)) • their angle from $360^\circ \times$ their period divided by 31.8 hours • evidence of 22.5° divided by their angle 433 or 434 or 430 ✓	Allow ecf from 01.2 if T calculated. Alternative route for MP1 and MP2 Two from: • calculates period • calculates time to move 22.5° (7155 s) • divides their time by their period condone answer that is not a whole number but rounds to 433 or 434	3	AO2

Total			10	
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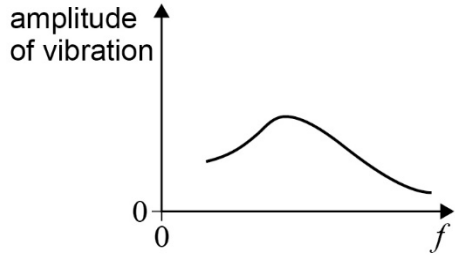
Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
07.1	Arrow drawn from bottom of blocks of wood to level with the centre of the pendulum bob ✓		1	AO4

Question	Answers	Additional comments/Guidelines	Mark	AO
07.2	Vertical line aligned with string in vertical position. Part of card should be behind and below the bob ✓ The bob is moving fastest here (so the uncertainty in judging the time when it passes the mark is smallest) ✓		2	1 × AO3 1 × AO4

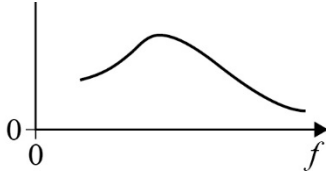
Question	Answers	Additional comments/Guidelines	Mark	AO
07.3	Any two from: ✓✓ • measure $10T$ and divide by 10 • repeat and average • ensure pendulum is viewed perpendicular to oscillations	Condone: • ensure that the pendulum is only allowed to swing through a small angle ($<15^\circ$) / pendulum is only displaced a small amount / small-amplitude oscillations • countdown method for timing	2	AO4

Question	Answers	Additional comments/Guidelines	Mark	AO
07.4	Recorded half oscillations instead of full oscillations ✓ $g \left(= \frac{39.1}{4} \right) = 9.78 \text{ (m s}^{-2}\text{)} \checkmark$		2	AO3
Total			7	

Question	Key	Answer	AO
08	C	0.37 m s^{-1}	AO1
09	D		AO1
10	A	$\frac{R}{4}$	AO1
11	D	9.0 nC	AO2
12	A	A s V^{-1}	AO1
13	C	0.08 3.6	AO2
14	D	8.37×10^{19}	AO2
15	B	$\ln(\text{number of decayed nuclei})$ against $\ln(\text{time})$	AO1
16	B	The kinetic energy of the particle is constant.	AO1

17	C	69°	AO2
18	C	$2A \quad 0.5f$	AO2
19	A	P and R arrive together, followed by Q .	AO1

8 A	AO2
95%	AO2
2.8 cm	AO2

09	D		AO1
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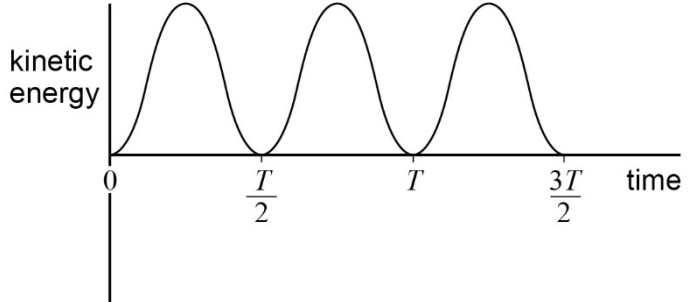
22	C	
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Total marks = 15

Question	Answers	Additional comments/Guidelines	Mark	AO
06.1	0.15 (s) ✓	$\frac{1}{4}T = \frac{1}{4} \times \frac{1}{1.7} = 0.147 \text{ s}$	1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
06.2	Use of $x = A \cos \omega t$ or $\omega = 2\pi f$ ✓ 0.20 (s) ✓	$\omega = 2\pi f = 2\pi \times 1.7 = \frac{17\pi}{5} = 10.68$ $x = A \cos \omega t$ $-1.8 = 3.6 \cos 10.68 \times t$ $t = 0.1961$	2	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
06.3	Any one from: ✓ • use of max speed = ωA and $E_k = \frac{1}{2}mv^2$ • use of maximum E_k or total energy = $\frac{1}{2}m\omega^2 A^2$ 0.051 (kg) ✓	ecf from 06.2 $\max E_k = \frac{1}{2}m\omega^2 A^2$ $m = \frac{2 \times E_k}{\omega^2 A^2} = \frac{2 \times 3.8 \times 10^{-3}}{10.68^2 \times (3.6 \times 10^{-2})^2} = 0.0514$	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
06.4	Graph shows maximum only at the halfway points between the scale markings and zero only at the scale markings ✓ Graph varies sinusoidally (by eye) between 0 and maximum ✓	 A line which is negative at any point scores 0 If the maxima noticeably vary by eye then do not award mp2	2	AO3
Total			7	

Question	Answers	Additional comments/Guidelines	Mark	AO
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11	D	π rad	AO1
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12	c	$2.2 \times 10^{-2} \text{ J}$	AO2
			AO3

Unknown Unknown - Unit 3 - Q06

question	marking guidance	mark	comments
6.1	acceleration is proportional to displacement ✓ acceleration is in opposite direction to displacement, or towards a fixed point, or towards the centre of oscillation ✓	2	
06.2	$\frac{25}{23} = 1.09$ seen ✓	1	
6.3	(use of $a = (2\pi f)^2 A$ gives) $a = (2\pi \times 1.09)^2 \times 76 \times 10^{-3}$ ✓ $= 3.5 \text{ m s}^{-2}$ ✓	2	
6.4	(use of $x = A \cos(2\pi ft)$ gives) $x = 76 \times 10^{-3} \cos(2\pi \times 1.09 \times 0.60)$ ✓ $= (-)4.3(1) \times 10^{-2} \text{ m}$ ✓ (43 mm) (use of $f = 1.1 \text{ Hz}$ gives $x = (-)4.0(7) \times 10^{-2} \text{ m}$ (41 mm))	2	
6.5	Energy always positive ✓ 4 complete oscillations shown ✓ $E_k = 0$ at $t = 0$ ✓	max 2	
6.6	Mention of resonance ✓ large amplitude of vibration of mass ✓ because driven close to the natural frequency ✓	3	
6.7	the system is now <u>damped</u> ✓ energy is transferred (to the water) so that amplitude is reduced ✓	2	

question	marking guidance	mark	comments
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